SAFEFLOW RAMS - TEST OUTPUT
Task: Excavation / Groundworks
Location: Residential access road, Bristol BS3
Status: Generated from fresh mobile plant / excavation image test

Image inputs:
- test-outputs/mobile-plant/images/excavator-trench.jpg
- test-outputs/mobile-plant/images/pipeline-trench.jpg
- test-outputs/mobile-plant/images/mobile-plant-grading.jpg

SCOPE OF WORKS

Excavate a short trench across an access road, expose and repair a damaged duct, and reinstate the granular sub-base. Operations involve a compact excavator, dumper, and vibrating plate compactor while maintaining pedestrian access and protecting nearby parked vehicles.

SITE OBSERVATIONS

The site is located in a residential area with close proximity to pedestrian and vehicular access. Excavation is performed near existing pipelines and services, suggesting potential underground utilities. Multiple pieces of heavy machinery are in use, indicating a need for caution around mobile plant operations. Environmental factors include exposure to dust and possible noise concerns.

CRITICAL RISK VALIDATION

Detected rules:
- Ground Disturbance / Underground Services (ground-disturbance-services)
- Live Traffic / Highway Interface (live-traffic-highways)
- Public Interface (public-interface)
- Overhead Services (overhead-services)
- Silica / Dust (silica-dust)
- Noise / Vibration (noise-vibration)
- COSHH Substances (coshh-substances)
- Plant / Pedestrian Interface (plant-pedestrian-interface)
- Machinery / Work Equipment (machinery-work-equipment)
- Mobile Plant (mobile-plant)
- Excavation Collapse / Ground Instability (excavation-collapse)
- Environmental Spill / Pollution (environmental-spill)
- Weather / Temperature Exposure (weather-exposure)

Validation gaps found:
- Environmental Spill / Pollution: Fuel, oil or material release causing environmental harm.

HAZARD REGISTER
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1. Underground services strike [High -> Medium]
   Those at risk: Site workers, nearby residents
   Initial: L4 x S4 | Residual: L2 x S3
   Controls: Review statutory utility plans and site service drawings. Perform CAT and Genny scans. Use permit to dig. Conduct trial holes and use hand digging near suspected services. Escalate if unidentified services are found.

2. Vehicle intrusion during live traffic interface [High -> Medium]
   Those at risk: Site workers, road users
   Initial: L4 x S5 | Residual: L2 x S3
   Controls: Implement temporary traffic management with cones and barriers. Use traffic management operatives. Ensure high-visibility clothing and work zone lighting.

3. Public interference with work areas [High -> Low]
   Those at risk: Site workers, public
   Initial: L3 x S4 | Residual: L2 x S2
   Controls: Use barriers, signage, and controlled access points. Maintain safe pedestrian routes. Use a marshal where public interaction cannot be segregated.

4. Contact with overhead services [High -> Medium]
   Those at risk: Site workers using machinery
   Initial: L3 x S5 | Residual: L2 x S3
   Controls: Identify overhead services and confirm exclusion distances. Use goalposts and height restrictors. Employ a spotter during operations.

5. Dust exposure [Medium -> Low]
   Those at risk: Site workers, nearby residents
   Initial: L3 x S3 | Residual: L1 x S2
   Controls: Use water suppression and RPE. Segregate dusty work areas and clean using wet methods.

6. Noise and vibration exposure [Medium -> Low]
   Those at risk: Site workers
   Initial: L3 x S4 | Residual: L1 x S2
   Controls: Assess HAVS and noise exposure, limit trigger time. Use low-vibration tools, rotate tasks, and provide hearing protection.

7. Collision between plant and pedestrians [High -> Medium]
   Those at risk: Site workers, pedestrians
   Initial: L4 x S5 | Residual: L2 x S3
   Controls: Use exclusion zones with barriers. Assign a banksman for guiding plant movements. Ensure plant pre-use checks.

8. Excavation collapse [High -> Low]
   Those at risk: Site workers
   Initial: L3 x S5 | Residual: L1 x S3
   Controls: Assess ground conditions, use shoring as needed. Keep spoil away from edges. Provide edge protection and inspect after weather events.

METHOD STATEMENT
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1. Review site plans and perform CAT scans. 2. Set up safe zones and traffic management. 3. Conduct site induction specific to the project. 4. Begin excavation using hand tools near identified services. 5. Use compact excavator to remove soil. 6. Expose and repair duct. 7. Refill trench with granular sub-base and compact. 8. Restore area and remove site equipment.

PPE

High-visibility clothing, Hard hats, Safety boots, Gloves, Hearing protection, Dust masks

WELFARE ARRANGEMENTS

Welfare facilities must include toilets, hand washing stations, and a rest area, in compliance with CDM 2015 Regulation 13.

ENVIRONMENTAL CONTROLS

Avoid ground contamination with spill kits and secure waste management. Monitor air quality for dust and noise levels.

COSHH ASSESSMENT

Review SDS/COSHH assessments for substances like fuel and oil on site to mitigate exposure risks.

REFUELLING PROCEDURE
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1. Move plant to designated refuelling area. 2. Turn off equipment. 3. Use spill kit under nozzle. 4. Replace fuel cap securely. 5. Clean up any spills immediately.

TRAINING REQUIREMENTS

- CSCS Card
- Excavator Operator Certification
- CAT and Genny Training
- Traffic Management Certification

EMERGENCY ARRANGEMENTS

Dial 999 in the event of an emergency. Nearest A&E is Southmead Hospital, Southmead Road, Westbury-on-Trym, Bristol BS10 5NB.

COMPETENCIES

All operatives must have relevant CSCS cards, NVQs in construction operations, and completed site safety inductions.

REFERENCES

- HSG47: Avoiding danger from underground services: https://www.hse.gov.uk/pubns/books/hsg47.htm
- HSG150: Health and safety in construction: https://www.hse.gov.uk/pubns/books/hsg150.htm
- COSHH Essentials - advice sheets: https://www.hse.gov.uk/coshh/essentials/direct-advice/index.htm
- HSG144: Safe use of vehicles on construction sites: https://www.hse.gov.uk/pubns/books/hsg144.htm
- Construction health and safety: https://www.hse.gov.uk/construction/index.htm